

# $\sigma$ : Sigmoid Modulation for Ultra High Resolution Diffusion

Bingxuan Zhao<sup>1 2</sup> Qing Zhou<sup>2</sup> Yu Wang<sup>2</sup> Chuang Yang<sup>2</sup> Qi Wang<sup>2</sup>

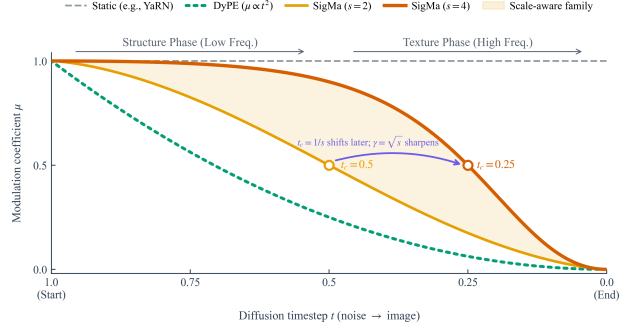
## Abstract

Diffusion Transformers (DiTs) can synthesize high-fidelity images, but training at ultra-high resolutions is expensive, making inference-time extrapolation essential. Existing methods are typically *scale-agnostic*, applying the same positional-modulation schedule regardless of target resolution. We show that this misses a scale-sensitive property of denoising: resizing shifts a fixed semantic pattern toward lower *normalized* spatial frequencies while enlarging the spatial support over which global structure must be coordinated. The latter delays structural lock-in at high resolutions, so schedules may either relax guidance before global layout has stabilized, causing *structural collapse*, or retain excessive intervention into late denoising, causing *textural degradation*. We introduce **SigMa** ( $\sigma$ ), a training-free framework that uses Sigmoid Modulation for *scale-adaptive* extrapolation through two scaling laws: *Decoupled Geometric Center Alignment* and *Iso-Variance Rate Adaptation*. Experiments show that SigMa reduces this mismatch, enabling training-free extrapolation up to 16 megapixels and achieving the best or competitive performance among the tested training-free extrapolation baselines. Code is available at [github.com/bxuanz/SigMa](https://github.com/bxuanz/SigMa).

## 1. Introduction

Diffusion Transformers (DiTs) (Peebles & Xie, 2023; Esser et al., 2024) have become a strong paradigm for generative modeling, combining the scalability of transformers with the stable training objectives of diffusion (Vaswani et al., 2017; Labs, 2024). Although this architecture enables high-fidelity synthesis, training DiTs at ultra-high resolutions (e.g.,  $4K^2$ ) remains computationally expensive due to the

<sup>1</sup>School of Computer Science, Northwestern Polytechnical University, Xi'an, China <sup>2</sup>School of Artificial Intelligence, OPTics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China. Correspondence to: Qi Wang <crabwq@gmail.com>.



**Figure 1. Breaking the Rigid Paradigm with SigMa. Spectral Dynamics:** Diffusion intrinsically transitions from a *Structure Phase* (Low Freq.) to a *Texture Phase* (High Freq.), a shift that is inherently *scale-dependent*. **The Rigid Trap:** Existing methods (dashed/dotted lines) ignore this dependency. By applying resolution-invariant schedules (e.g.,  $\mu \propto t^2$ ), they can force a **pre-mature transition** to texture refinement at large scales, causing structural collapse. **SigMa (Ours):** The solid curves visualize the final schedule  $\mu_d(t) = \sigma(\sqrt{s_d}[\ell(t) - \ell(s_d^{-1})])$ . Larger scales move the center  $t_c = 1/s_d$  toward later, cleaner timesteps while increasing the rate  $\gamma = \sqrt{s_d}$ , securing global structure before handing over to detail refinement.

quadratic complexity of self-attention (Dosovitskiy et al., 2020; Chen et al., 2024). As a result, the field relies heavily on **training-free extrapolation** techniques, such as Position Interpolation (PI) (Chen et al., 2023), NTK-aware interpolation (bloc97, 2023), and YaRN (Peng et al., 2024), to extend pre-trained capabilities to unseen resolutions.

Despite their strong empirical performance, existing extrapolation paradigms (bloc97, 2023; Peng et al., 2024; Issachar et al., 2025) typically follow a *scale-agnostic premise*, treating the denoising dynamics identically regardless of target resolution. As illustrated in Figure 1, this choice leads to two characteristic failure modes. Traditional methods (e.g., YaRN, grey dashed line) rely on *static schedules* that apply a uniform modulation throughout the process, effectively ignoring the **spectral evolution** of diffusion, i.e., the progression from low-frequency structural construction to high-frequency texture refinement (Rombach et al., 2022; Karras et al., 2022). While DyPE (Issachar et al., 2025) introduces time-dependent modulation, it remains constrained by a *rigid heuristic* (e.g.,  $\mu \propto t^2$ ) that is invariant across scales. We argue that both static and rigid-dynamic paradigms fail to account for the scale-dependency of the temporal win-

dows required for these phases: larger resolutions require a longer structural phase to establish global coherence before the model can shift toward fine-grained details.

This mismatch between rigid modulation schedules and increasing spatial extent creates an evolution gap and can lead to spectral misalignment. The denoising trajectory is not static, but a progressive transition from low-frequency global structures to high-frequency details (Rombach et al., 2022; Choi et al., 2022). Recent work on noise schedule shifting (Hooeboom et al., 2023; Lin et al., 2024) further suggests that the effective signal-to-noise requirements change with resolution, making schedules tuned for smaller scales less suitable for ultra-high-resolution synthesis. At these larger scales, rigid decay schedules may shift into texture refinement before the global structure is sufficiently secured, resulting in structural collapse and object duplication, a spatial analogue of the attention sinks observed in long-context transformers (Xiao et al., 2024).

SigMa addresses this mismatch through two scale-aware design rules: *Decoupled Geometric Center Alignment* delays the transition timing to match the resolution-dependent structural lock-in point, while *Iso-Variance Rate Adaptation* adjusts the transition sharpness according to the growth of spatial feature variance. Crucially, this formulation enables **axis-decoupled calibration**, allowing SigMa to synchronize the denoising trajectory for height and width independently. By aligning positional intervention with the geometric demands of the target resolution, SigMa preserves both structural integrity and fine-grained photorealism even under extreme aspect ratios.

Our contributions are summarized as follows:

- **Scale-Dependent Insight:** We identify the scale dependence of spectral evolution in DiTs and show why scale-agnostic heuristics can lead to spectral misalignment.
- **SigMa Framework:** We propose SigMa, a training-free paradigm that uses **axis-decoupled** Sigmoid Modulation to adaptively calibrate transition timing and velocity.
- **Strong Empirical Performance:** Extensive experiments show that SigMa reduces extrapolation artifacts, supports high-fidelity synthesis at resolutions up to 16 megapixels, and achieves best or competitive results among strong baselines, particularly in challenging panoramic regimes.

The remainder of the paper is organized as follows. Section 2 reviews RoPE extrapolation methods and their limitations. Section 3 develops the scale-dependent evolution-gap view and introduces SigMa. Section 4 evaluates the method

on high-resolution, panoramic, and ablation benchmarks, while the appendix provides integration details, additional quantitative and qualitative results, and transfer evidence on Qwen-Image.

## 2. Related Work

### 2.1. Rotary Positional Embeddings

Since the standard Transformer block is permutation-equivariant, a positional encoding mechanism is essential to model the strong spatial dependencies inherent in natural images (LeCun & Bengio, 1998). Early approaches relied on fixed sinusoidal embeddings (Vaswani et al., 2017; Dosovitskiy et al., 2020), learned absolute embeddings (Devlin et al., 2019; Radford et al., 2019), or relative positional embeddings (Press et al., 2021).

More recently, Rotary Positional Embeddings (RoPE) (Su et al., 2024; Touvron et al., 2023; Heo et al., 2024) have emerged as a superior alternative. In the context of Vision Transformers, RoPE is typically applied as *Axial RoPE*, decomposing the 2D spatial grid into orthogonal 1D components. RoPE encodes the position index  $m$  by rotating the feature vector in a complex vector space. Specifically, for a dimension index  $d \in [0, D/2)$  where  $D$  is the head dimension, the rotation frequency  $\theta_d$  is derived from a geometric sequence:

$$\theta_d = \theta_{base}^{-2d/D}, \quad (1)$$

where  $\theta_{base}$  is a hyper-parameter (e.g., 10,000). While RoPE effectively preserves relative positional information within the training resolution (length  $L$ ), directly extrapolating it to a longer context  $L' > L$  during inference introduces out-of-distribution rotation angles. This discrepancy disrupts the attention mechanism, leading to *attention entropy collapse* and generation failure.

### 2.2. Static Inference-time Extrapolation

To enable high-resolution synthesis without retraining, various *static* extrapolation methods have been proposed. These methods adjust the positional mapping  $m \mapsto g(m)$  or the frequency response  $\theta_d \mapsto h(\theta_d)$  using a fixed scale factor  $s = L'/L$  throughout the entire sampling process.

**Position Interpolation (PI).** PI (Chen et al., 2023) linearly scales the position indices to match the training domain boundaries. The mapping is defined as:

$$g(m) = m/s, \quad h(\theta_d) = \theta_d. \quad (2)$$

While PI prevents domain shifts, the uniform compression of coordinates inevitably squeezes the entire frequency spectrum. This reduces the upper bound of representable frequencies, resulting in the loss of fine details and blurry high-frequency textures.

**NTK-Aware Interpolation.** To preserve high-frequency fidelity, NTK-Aware interpolation (bloc97, 2023) applies a non-uniform scaling based on Neural Tangent Kernel theory. It aims to keep high-frequency components unchanged while interpolating low-frequency components:

$$g(m) = m, \quad h(\theta_d) = \frac{\theta_d}{s^{2d/(D-2)}}. \quad (3)$$

This transformation ensures that fine details (high frequencies) are generated using the original representation capacity, while global structures (low frequencies) are adapted to the extended context window.

**YaRN.** Yet another RoPE extension (YaRN) (Peng et al., 2024) further refines this by introducing the *NTK-by-parts* strategy. It divides the frequency spectrum into three bands—interpolating low frequencies, extrapolating high frequencies, and smoothing the transition for middle frequencies via a ramp function  $\rho(r)$ . This ramp function is determined by two static threshold parameters,  $\beta_{min}$  and  $\beta_{max}$  (originally denoted as  $\alpha$  and  $\beta$ ):

$$\rho(r) = \begin{cases} 0, & r < \beta_{min} \\ \frac{r - \beta_{min}}{\beta_{max} - \beta_{min}}, & \beta_{min} \leq r \leq \beta_{max} \\ 1, & r > \beta_{max} \end{cases} \quad (4)$$

where  $r = L/\lambda_d$  denotes the wavelength ratio relative to the context size. The frequency response is then modulated as:

$$h(\theta_d) = (1 - \rho(r)) \frac{\theta_d}{s} + \rho(r) \theta_d. \quad (5)$$

Furthermore, to counteract the attention entropy increase caused by the extended context, YaRN introduces a temperature scaling factor  $\sqrt{\tau}$  to the attention logits. By carefully balancing frequency preservation and attention distribution, YaRN serves as a robust baseline for static resolution adaptation.

### 2.3. Dynamic Position Extrapolation

To overcome the limitations of static transformations, DyPE (Issachar et al., 2025) introduces a plug-and-play framework that imparts time-dependency to existing extrapolation methods. It modulates the intensity of positional intervention using a power-law scheduler:

$$\mu_{\text{DyPE}}(t) = \lambda_s \cdot t^{\lambda_t}, \quad (6)$$

where  $t$  is the normalized denoising timestep. By integrating this scheduler into static formulae (e.g., modulating YaRN’s thresholds), DyPE enforces structural constraints early in the process and relaxes them for texture refinement.

However, the control mechanism of DyPE remains *scale-agnostic*. The decay characteristic is governed by a fixed

hyperparameter  $\lambda_t$  that does not explicitly adapt to the target resolution. Consequently, DyPE applies an identical temporal schedule regardless of whether the scale is  $2\times$  or  $16\times$ . This rigidity can fail to synchronize with the scale-dependent shift in the preferred structural formation window, leading to misalignment at extreme resolutions.

## 3. Method

We introduce **SigMa**, a training-free framework designed to synchronize the modulation of positional embeddings with the inherent *spectral evolution* of the diffusion process. To rectify the **spectral misalignment** identified in Sec. 1, our approach builds on a mechanism-oriented analysis of how resolution scaling shifts the temporal boundaries between global structure formation and local texture refinement.

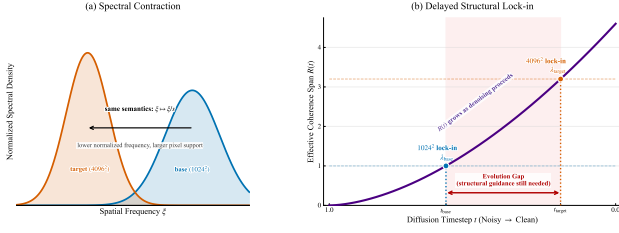
### 3.1. The Scale-Dependence of Spectral Evolution

As motivated in Sec. 1, the denoising dynamics in Diffusion Transformers (DiTs) is not a monolithic progression but a *phased transition* from low-frequency global structures to high-frequency textures (Choi et al., 2022; Rombach et al., 2022). We argue that existing extrapolation methods, such as PI (Chen et al., 2023) or DyPE (Issachar et al., 2025), encounter bottlenecks because they treat this transition boundary as a scale-agnostic constant, ignoring its intrinsic dependence on the target resolution dimensions and the resulting shift in the structural formation window.

**Spectral Contraction and Semantic Displacement.** We first characterize the impact of spatial scaling on the image power spectrum. Let  $s = (s_h, s_w)$  denote the resolution scaling factors, where  $s_h = H_{target}/H_{base}$  and  $s_w = W_{target}/W_{base}$ . According to the Fourier Similarity Theorem (Debnath & Bhatta, 2016), if a baseline image  $x(u)$  is scaled to form  $x'(u) = x(u_h/s_h, u_w/s_w)$ , its Fourier transform  $\mathcal{F}\{x'\}$  satisfies:

$$\mathcal{F}\{x'\}(\xi_h, \xi_w) = |s_h s_w| \cdot \mathcal{F}\{x\}(s_h \xi_h, s_w \xi_w). \quad (7)$$

Thus, after upscaling, the same continuous semantic pattern is represented at a lower *normalized* spatial frequency on the enlarged grid. This observation is important but insufficient on its own: frequency coordinates describe where signal energy lies in Fourier space, whereas structural stabilization depends on the spatial support over which the denoiser must coordinate consistent predictions. After scaling, the same object boundary, part arrangement, or horizon line spans more pixels, so the model must establish coherence over a larger geometric extent even though the normalized frequency is lower. This distinction is the key to reconciling spectral contraction with delayed structural convergence.



**Figure 2. Scale-Dependent Spectral Evolution.** (a) **Spectral contraction:** Upscaling by factor  $s$  compresses the normalized frequency coordinates of the same semantic pattern according to the Fourier Similarity Theorem. (b) **The Evolution Gap:**  $R(t)$  denotes the effective coherence span at timestep  $t$ . As denoising proceeds and SNR increases,  $R(t)$  grows, shifting the structural lock-in threshold from  $\lambda_{\text{base}}$  at  $t_{\text{base}}$  to the larger  $\lambda_{\text{target}}$  at a later, cleaner step  $t_{\text{target}} < t_{\text{base}}$ . Prematurely withdrawing guidance inside this gap leaves the high-resolution model with an unconverged global layout, causing structural collapse.

**The Evolution Gap and SNR-Dependent Lock-in.** A key property of diffusion models is that denoising proceeds from coarse, low-frequency structure to fine-grained, high-frequency detail (Karras et al., 2022; Hoogeboom et al., 2023). To make the scale effect precise, we define  $R(t)$  as the *effective coherence span* at normalized timestep  $t$ , where  $t = 1$  corresponds to the noise-dominated start and  $t = 0$  to the clean end. Intuitively,  $R(t)$  is the largest spatial extent over which the denoiser can produce mutually consistent structure at that noise level. The standard coarse-to-fine view implies that this reliable span grows along the sampling trajectory as  $t$  moves from 1 to 0 and SNR rises.

**Proposition 3.1** (Delayed Structural Lock-in). *Consider a layout-critical structure whose span along axis  $d$  is  $\lambda_{\text{base}}$  at the base resolution and  $\lambda_{\text{target}} = s_d \lambda_{\text{base}}$  after scaling by  $s_d \geq 1$ . Define its lock-in time as*

$$t_{\text{lock}}(s_d) = \sup \{t \in [0, 1] : R(t) \geq s_d \lambda_{\text{base}}\}. \quad (8)$$

Then  $t_{\text{lock}}(s_d) \leq t_{\text{lock}}(1)$ .

*Proof.* Because  $s_d \lambda_{\text{base}} \geq \lambda_{\text{base}}$ , the set of timesteps satisfying the scaled lock-in condition is contained in the corresponding base-resolution set. Taking suprema preserves the inclusion order, so the scaled lock-in time cannot occur earlier than the base-resolution lock-in time.

Proposition 3.1 captures the central failure mode of scale-agnostic scheduling: the timestep at which large-scale structure becomes reliable shifts toward cleaner states as the target span increases. We call the interval  $[t_{\text{target}}, t_{\text{base}}]$ , with  $t_{\text{target}} = t_{\text{lock}}(s_d)$  and  $t_{\text{base}} = t_{\text{lock}}(1)$ , the **Evolution Gap**. Inside this interval, a baseline schedule would already start relaxing positional intervention, while the high-resolution trajectory still requires structural support. If guidance is withdrawn in this gap, the model is forced to refine local

details on top of an unconverged global scaffold, leading to structural collapse and geometric repetition.

### 3.2. SigMa: Scale-Adaptive Calibration Framework

To bridge the evolution gap identified in Sec. 3.1, we introduce **SigMa** (Sigmoid Modulation Adaptation). Unlike rigid power-law heuristics that ignore the scale-dependence of spectral evolution, SigMa reconceptualizes extrapolation as a **scale-adaptive phase transition**. Our objective is to orchestrate axis-specific modulation schedules  $\mu_d(t)$  whose centers and widths track the delayed structural lock-in point of ultra-high-resolution inputs as closely as possible.

**The General Formulation.** For each extrapolated spatial axis  $d \in \{w, h\}$  with  $s_d > 1$ , we implement a parameterized spectral gate  $\mu_d(t)$  to modulate the intensity of positional intervention during denoising ( $t : 1 \rightarrow 0$ ):

$$\begin{aligned} \mu_d(t) &= \sigma(\gamma_d[\ell(t) - \ell(t_{c,d})]), \\ \ell(t) &= \log \frac{t}{1-t}, \quad t \in (0, 1), \\ \sigma(x) &= \frac{1}{1 + \exp(-x)}. \end{aligned} \quad (9)$$

Equation 9 gives the abstract two-parameter gate. Instantiating it with the scale laws derived below gives the concrete SigMa schedule used throughout the paper:

$$\mu_d(t) = \sigma(\sqrt{s_d}[\ell(t) - \ell(s_d^{-1})]), \quad s_d > 1. \quad (10)$$

This reparameterization keeps the open-interval endpoint behavior simple:  $\lim_{t \rightarrow 0^+} \mu_d(t) = 0$ ,  $\lim_{t \rightarrow 1^-} \mu_d(t) = 1$ , and  $\mu_d(t_{c,d}) = 1/2$ , without any extra normalization term or compression hyperparameter. Here,  $\mu_d(t) \in [0, 1]$  governs the spectral hand-off on axis  $d$ : larger values retain stronger structural guidance, while smaller values relax the positional intervention and expose more native high-frequency detail. The sigmoid gate matches the desired structure-to-texture hand-off with few degrees of freedom: it is smoother than a hard switch, naturally bounded unlike a linear ramp, and more flexible than a fixed power law such as DyPE’s  $\mu \propto t^2$  because its center  $t_{c,d}$  and rate  $\gamma_d$  are separately controllable. The log-odds time  $\ell(t)$  further gives an exact midpoint interpretation while preserving simple limiting behavior on  $t \in (0, 1)$ . For a non-extrapolated axis ( $s_d = 1$ ), no positional rescaling is required, so we do not evaluate the boundary case  $t_{c,d} = 1$  in Eq. 9; instead, we simply keep  $\hat{s}_d(t) = 1$ . Because modern DiTs (e.g., FLUX (Labs, 2024)) employ Axial RoPE and can be extrapolated anisotropically, we treat the delay in structural convergence separately along the height and width axes. We therefore propose an **Axis-Decoupled Calibration** strategy. Let  $s_d$  denote the scale factor for a specific dimension  $d \in \{w, h\}$ . We calibrate the transition timing  $t_c$  and velocity  $\gamma$  independently for each axis.



"A silhouetted pagoda stands against a large red moon, surrounded by dark mountains and trees..."

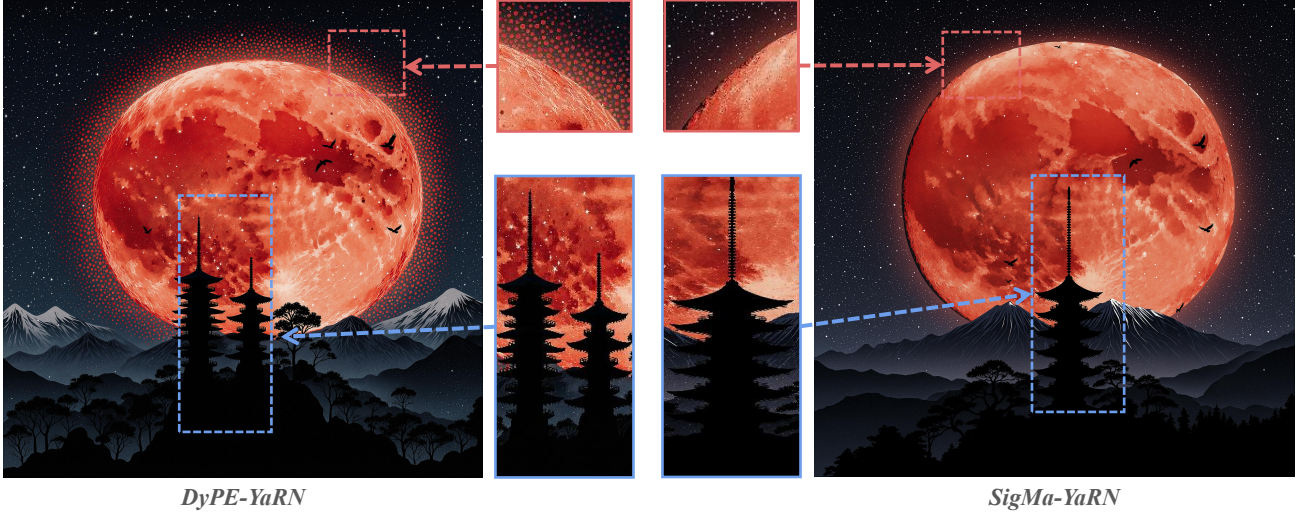


Figure 3. **Visual comparison at  $4\times$  extrapolation ( $4096^2$ ).** Comparison between DyPE-YaRN (Left) and SigMa-YaRN (Right) for the same prompt. The magnified regions show two complementary differences: DyPE-YaRN preserves the coarse red-moon composition but introduces noisy boundary artifacts around the moon and less stable pagoda silhouettes, whereas SigMa-YaRN keeps a cleaner global scaffold with sharper local contours. This example is consistent with the evolution-gap analysis: scale-adaptive timing maintains structural guidance longer at high resolution, while the later relaxation of positional intervention recovers cleaner high-frequency detail.

**Mechanism I: Decoupled Geometric Center Alignment ( $t_c$ ).** This mechanism addresses the timing of the phase transition. Proposition 3.1 fixes the direction of the shift: a larger target span should postpone the transition on the normalized denoising horizon. To determine a closed-form center without adding a tunable exponent, we use a scale-consistency condition.

**Proposition 3.2** (Scale-Consistent Center). *If geometrically similar extrapolation problems share the same transition center in the scale-normalized phase variable  $u = s_d t$ , and the native-resolution normalization is  $t_c(1) = 1$ , then*

$$t_c(s_d) = \frac{1}{s_d}, \quad \text{for extrapolated axes } s_d > 1. \quad (11)$$

*Proof.* Let  $B_c(s_d) = s_d t_c(s_d)$  be the dimensionless center budget in the variable  $u = s_d t$ . Scale consistency requires  $B_c(s_d)$  to be invariant across geometrically similar extrapolation problems. Since  $t_c(1) = 1$ , we have  $B_c(1) = 1$  and therefore  $B_c(s_d) \equiv 1$ , which gives  $t_c(s_d) = 1/s_d$ .

*Interpretation:* Under the convention  $t : 1 \rightarrow 0$ , larger scales receive smaller  $t_c$  and therefore switch later. Differentiating the center law gives  $dt_c/ds_d = -1/s_d^2$ , so the center drift itself becomes less sensitive at larger scales. The log-odds sigmoid in Eq. 9 keeps this center interpretation explicit:  $t_c$  marks the half-activation point, so moving  $t_c$  toward cleaner timesteps directly prolongs structural guidance. For a  $4096 \times 1024$  image ( $s_w = 4, s_h = 1$ ), SigMa assigns  $t_c = 0.25$  for the width axis to prolong structural guidance.

The height axis is already native when  $s_h = 1$ , so its effective scale remains  $\hat{s}_h(t) = 1$ . Figure 5(a) and Table 2a show that nearby alternatives such as  $1/s^2$  and  $1/\sqrt{s}$  are weaker in our tested setting, supporting the inverse-linear law as the scale-consistent center rule for this regime.

**Mechanism II: Iso-Variance Rate Adaptation ( $\gamma$ ).** This mechanism regulates the velocity, or sharpness, of the phase transition. Let  $\Delta\tau_{\text{window}} \propto 1/\gamma$  denote the temporal width of the mixed regime in log-odds time, where  $\tau = \ell(t)$ . Along an elongated axis, this mixed regime is driven by aggregated features over more tokens. For finite-correlation spatial responses, the standard deviation of the aggregated signal scales as  $\Sigma_{\text{signal}}(s_d) \propto \sqrt{s_d}$ . The parameter  $\gamma_d$  is therefore the scale-dependent term that controls the concentration of the crossover around  $t_{c,d}$ .

**Proposition 3.3** (Iso-Variance Rate). *If the crossover width in log-odds time satisfies  $\Delta\tau_{\text{window}} \propto 1/\gamma$  and the aggregated signal spread obeys  $\Sigma_{\text{signal}}(s_d) \propto \sqrt{s_d}$ , then preserving the standardized crossover width across scales yields*

$$\gamma(s_d) = \sqrt{s_d}, \quad \text{for extrapolated axes } s_d > 1. \quad (12)$$

*Proof.* The iso-variance criterion keeps  $\Delta\tau_{\text{window}} \cdot \Sigma_{\text{signal}}(s_d)$  approximately constant. Substituting  $\Delta\tau_{\text{window}} \propto 1/\gamma(s_d)$  and  $\Sigma_{\text{signal}}(s_d) \propto \sqrt{s_d}$  gives  $1/\gamma(s_d) \cdot \sqrt{s_d} \approx \text{const}$ . Normalizing at the base scale fixes the constant, yielding  $\gamma(s_d) = \sqrt{s_d}$ .

*Interpretation:* As the elongated axis grows, the transition becomes sharper just enough to offset the broader feature

"A large planet with prominent rings occupies the foreground against a backdrop of deep red space, while two..."



*Flux*

*YaRN*

*DyPE-YaRN*

*SigMa-YaRN*

"A blue-skinned character with pointed ears and glasses sits at a wooden desk cluttered with maps and navigational..."



*Flux*

*NTK-aware*

*DyPE-NTK-aware*

*SigMa-NTK-aware*

Figure 4. **Qualitative comparison at 4096<sup>2</sup>**. Visual results of **SigMa** against baselines on *YaRN-based* (Top) and *NTK-based* (Bottom) methods. Within each row, columns show direct FLUX inference, the static extrapolator, its DyPE variant, and the corresponding SigMa variant.

statistics induced by more spatial positions. Appendix A details this finite-correlation aggregation argument, and Table 4 shows that the measured hidden-state standard deviation in FLUX.1-dev closely follows the predicted  $\sqrt{s_d}$  law.

**Corollary 3.4** (Concrete SigMa Gate). *Combining Propositions 3.2 and 3.3 with Eq. 9 gives the schedule used in all experiments:*

$$\mu_d(t) = \sigma(\sqrt{s_d}[\ell(t) - \ell(s_d^{-1})]), \quad s_d > 1.$$

### 3.3. Unified Integration Strategy

SigMa functions as a plug-in dynamics module compatible with mainstream positional extrapolation methods. We use Dynamic Parameter Injection to render static baselines phase-aware without altering their underlying kernels.

Specifically, we inject the axis-specific spectral gate  $\mu_d(t)$  into the baseline’s control parameters to modulate their effective operating range. For global scaling methods (e.g., PI, NTK), this manifests as a *Time-Varying Effective Scale*  $\hat{s}_d(t) = s_d^{\mu_d(t)}$ ; for band-partitioning methods (e.g., YaRN), it acts as a *Dynamic Boundary Gate* on the ramp thresholds. We provide the exact mathematical formulations for

integrating SigMa into these paradigms in Appendix B.

## 4. Experiments

We evaluate SigMa on high-resolution generation, balancing global structural coherence with fine-grained textural fidelity. Using the FLUX.1-dev (Labs, 2024) backbone, our evaluation proceeds in three stages: (1) **comparative benchmarking** against strong training-free RoPE extrapolation baselines; (2) a controlled **ablation study** assessing the empirical support for our core mechanisms (Sec. 4.3); and (3) a test on **extreme aspect ratios** to evaluate axis-decoupled robustness (Sec. 4.4). Appendix C further complements these results with portrait and intermediate-scale evaluations, and Appendix C.2 reports transfer evidence on Qwen-Image.

### 4.1. Experimental Setup

**Implementation Details.** SigMa is deployed as a training-free adapter for FLUX.1-dev (Labs, 2024) with no per-resolution search. We utilize an Euler-flow solver (28 steps, guidance 3.5) across a spectrum of resolutions, from standard squares (3072<sup>2</sup>, 4096<sup>2</sup>) to panoramic layouts (4 : 1). In contrast to tunable dynamic schedules whose decay shape is

**Table 1. Quantitative Results on High-Resolution Synthesis.** Performance comparison on DrawBench and Aesthetic-4K across two resolutions per row. We report CLIPScore (CLIP), HPSv2 (HPS), and Aesthetic Score (Aes) for DrawBench, supplemented by FID for Aesthetic-4K. All methods are built on FLUX.

| Method                    | Landscape Extrapolation (3072 $\times$ 2048) |                |                  |                 |                |                  |                  | Square Extrapolation (4 $\times$ Scale, 4096 <sup>2</sup> ) |                |                  |                 |                |                  |                  |
|---------------------------|----------------------------------------------|----------------|------------------|-----------------|----------------|------------------|------------------|-------------------------------------------------------------|----------------|------------------|-----------------|----------------|------------------|------------------|
|                           | DrawBench                                    |                |                  | Aesthetic-4K    |                |                  |                  | DrawBench                                                   |                |                  | Aesthetic-4K    |                |                  |                  |
|                           | CLIP $\uparrow$                              | Aes $\uparrow$ | HPSv2 $\uparrow$ | CLIP $\uparrow$ | Aes $\uparrow$ | HPSv2 $\uparrow$ | FID $\downarrow$ | CLIP $\uparrow$                                             | Aes $\uparrow$ | HPSv2 $\uparrow$ | CLIP $\uparrow$ | Aes $\uparrow$ | HPSv2 $\uparrow$ | FID $\downarrow$ |
| <i>Baseline</i>           |                                              |                |                  |                 |                |                  |                  |                                                             |                |                  |                 |                |                  |                  |
| FLUX.1-dev                | 26.43                                        | 5.32           | 0.241            | 28.61           | 6.54           | 0.267            | 152.32           | 19.99                                                       | 3.81           | 0.139            | 25.70           | 5.42           | 0.200            | 197.21           |
| <i>NTK-based Methods</i>  |                                              |                |                  |                 |                |                  |                  |                                                             |                |                  |                 |                |                  |                  |
| NTK-aware                 | 27.45                                        | 5.52           | 0.253            | 28.86           | 6.62           | 0.275            | 150.11           | 17.80                                                       | 3.59           | 0.118            | 25.35           | 5.62           | 0.197            | 204.56           |
| DyPE-NTK                  | 27.67                                        | 5.66           | 0.261            | 29.02           | 6.69           | <b>0.281</b>     | 149.77           | 22.23                                                       | 4.43           | 0.177            | 27.73           | 6.38           | 0.254            | <b>184.99</b>    |
| <b>SigMa-NTK</b>          | <b>27.69</b>                                 | <b>5.71</b>    | <b>0.264</b>     | <b>29.10</b>    | <b>6.72</b>    | <b>0.281</b>     | <b>148.76</b>    | <b>22.29</b>                                                | <b>4.46</b>    | <b>0.179</b>     | <b>27.77</b>    | <b>6.41</b>    | <b>0.257</b>     | 185.35           |
| <i>YaRN-based Methods</i> |                                              |                |                  |                 |                |                  |                  |                                                             |                |                  |                 |                |                  |                  |
| YaRN                      | 27.47                                        | 5.67           | 0.264            | 29.19           | 6.79           | 0.282            | 151.21           | 25.78                                                       | 4.98           | 0.222            | 28.11           | 6.40           | 0.261            | 194.76           |
| DyPE-YaRN                 | 27.89                                        | 5.73           | 0.270            | 29.38           | 6.82           | 0.287            | 150.33           | 26.88                                                       | 5.24           | 0.244            | 28.72           | 6.63           | 0.279            | 188.53           |
| <b>SigMa-YaRN</b>         | <b>27.91</b>                                 | <b>5.76</b>    | <b>0.271</b>     | <b>29.46</b>    | <b>6.88</b>    | <b>0.288</b>     | <b>149.98</b>    | <b>26.90</b>                                                | <b>5.36</b>    | <b>0.247</b>     | <b>29.23</b>    | <b>6.71</b>    | <b>0.287</b>     | <b>184.73</b>    |

selected heuristically, SigMa analytically calibrates modulation from the target resolution and adds only scalar schedule updates during sampling.

**Baselines.** We benchmark against three paradigms corresponding to the failure modes analyzed in Sec. 1:

- Baseline: Direct inference with FLUX.1-dev (Labs, 2024), quantifying the lower-bound degradation.
- Static Modulation (NTK (bloc97, 2023), YaRN (Peng et al., 2024)): Time-invariant interpolation methods. These apply constant frequency compression, neglecting the temporal shift required for coarse-to-fine formation.
- Rigid Dynamic Modulation (DyPE (Issachar et al., 2025)): A time-variant strategy using a power-law schedule  $\mu_{\text{DyPE}}(t) = \lambda_s \cdot t^{\lambda_t}$ . We set  $\lambda_s = 2, \lambda_t = 2$  (quadratic decay). However, this method remains *scale-agnostic*, applying an identical decay curve regardless of the target resolution.

**Benchmarks & Metrics.** We employ a dual-benchmark protocol: DrawBench (Saharia et al., 2022) (200 prompts) to assess compositional alignment, and the Aesthetic-Eval@4096 subset of Aesthetic-4K (Zhang et al., 2025) (195 pairs, downsampled references) for ultra-high-resolution fidelity. We report CLIP-Score (Hessel et al., 2022) and Aesthetic Score (Schuhmann et al., 2022), while prioritizing HPSv2.1 (Wu et al., 2023) for human preference alignment. FID (Heusel et al., 2017) is reported for completeness despite sample size limitations.

## 4.2. Comparative Analysis

We systematically evaluate SigMa against the baseline categories defined in Sec. 4.1. As summarized in Table 1 and visualized in Figures 3 and 4, our empirical results indicate

that SigMa’s scale-adaptive calibration reduces the failure modes observed in both static and rigid dynamic paradigms. We provide further experimental analysis and extensive qualitative results in Appendix C to substantiate these findings.

### Improving Global Structural Coherence over Static

**Baselines.** Our evaluation shows that Direct Inference (FLUX.1-dev) can exhibit substantial geometric collapse when extrapolating to unseen resolutions. As illustrated in Figure 4 (col. 1), this appears as unrealistic lighting in planetary scenes or disjointed image layering in portrait generation. Quantitatively, this degradation is pervasive across the 4 $\times$  benchmarks but is most pronounced in compositional tasks, captured by an HPSv2 score of 0.139 on DrawBench (Table 1).

While static methods (NTK-aware, YaRN) attempt to mitigate this by rescaling frequencies, their time-invariant nature can still induce semantic repetition. For instance, the YaRN baseline (Figure 4, row 1) incorrectly renders multiple overlapping planetary rings, while NTK-aware (row 2) hallucinates subject duplication (creating two individuals instead of one). These artifacts are consistent with the frequency-to-time misalignment described by the evolution-gap view. In contrast, SigMa reduces these low-frequency artifacts by dynamically calibrating the spectrum. Visually, this improves the plausibility of the planetary rings and the structural integrity of the single-subject portrait. Quantitatively, SigMa-NTK and SigMa-YaRN consistently outperform their static counterparts across both DrawBench and Aesthetic-4K subsets.

### Improving Fine-Grained Fidelity over Rigid Dynamics.

While rigid dynamic methods (e.g., DyPE) mitigate the geometric repetition seen in static baselines by introducing



Table 2. **Ablation of Scaling Laws on Aesthetic-4K** (4096<sup>2</sup>). We conduct a controlled ablation to isolate the impact of Geometric Center ( $t_c$ ) and Dynamic Rate ( $\gamma$ ) within the YaRN framework.

| (a) Impact of Geometric Center ( $t_c$ ) |                        |                     |                           |                   | (b) Impact of Dynamic Rate ( $\gamma$ ) |                           |                  |                |                        |
|------------------------------------------|------------------------|---------------------|---------------------------|-------------------|-----------------------------------------|---------------------------|------------------|----------------|------------------------|
| Metric                                   | Const<br>( $t_c = 1$ ) | Late<br>( $1/s^2$ ) | Early<br>( $1/\sqrt{s}$ ) | Ours<br>( $1/s$ ) | Metric                                  | Const<br>( $\gamma = 1$ ) | Inv<br>( $1/s$ ) | Lin<br>( $s$ ) | Ours<br>( $\sqrt{s}$ ) |
| CLIP $\uparrow$                          | 28.05                  | 27.42               | 28.89                     | <b>29.23</b>      | CLIP $\uparrow$                         | 28.65                     | 27.60            | 28.75          | <b>29.23</b>           |
| Aes. $\uparrow$                          | 6.15                   | 6.01                | 6.52                      | <b>6.71</b>       | Aes. $\uparrow$                         | 6.38                      | 6.08             | 6.45           | <b>6.71</b>            |
| HPSv2 $\uparrow$                         | 0.248                  | 0.241               | 0.274                     | <b>0.287</b>      | HPSv2 $\uparrow$                        | 0.262                     | 0.245            | 0.269          | <b>0.287</b>           |
| FID $\downarrow$                         | 199.50                 | 208.30              | 189.10                    | <b>184.73</b>     | FID $\downarrow$                        | 194.20                    | 205.10           | 191.40         | <b>184.73</b>          |

time-variance, their limitation lies in their *scale-agnostic* nature. By adhering to a fixed power-law schedule ( $\mu \propto t^2$ ) regardless of target resolution or aspect ratio, these methods can impose a *one-size-fits-all* frequency compression that may misalign with the model’s denoising rate at extreme scales. Visually, this misalignment appears as visible degradation in textural fidelity. As shown in Figure 4 (row 2), although DyPE reduces the subject duplication seen in the static baseline, it still suffers from structural deformities, such as the manifestation of supernumerary limbs (three hands). Furthermore, it fails to maintain semantic fidelity for background elements, resulting in blurred or poorly defined objects like the map and compass.

In contrast, **SigMa** bridges the evolution gap through scale-adaptive calibration, helping maintain structural guidance until a stable foundation is secured. This improves the anatomical consistency seen in other methods while preserving high-fidelity skin textures and more coherent environmental lighting. Quantitatively, SigMa generally improves CLIP, Aesthetic Score, and HPSv2 over its DyPE counterpart while remaining competitive on FID, with the clearest gains appearing in the YaRN-based and panoramic settings.

### 4.3. Ablation Study

To empirically assess the core mechanisms in Sec. 3.2, we conduct a controlled decomposition on the Aesthetic-4K benchmark (4096<sup>2</sup>). We isolate the effects of the Decoupled Geometric Center ( $t_c$ ) and Iso-Variance Rate Adaptation ( $\gamma$ ). We visualize the overarching performance trends in Figure 5, supplemented by Table 2, which provides the exact quantitative metrics for the controlled comparison.

**Empirical Support for the Scale-Equivariant Center** ( $t_c = 1/s$ ). Figure 5(a) shows a clear contrast in structural integrity. A critical observation is the performance collapse of the overly late center-shrinkage candidate ( $1/s^2$ ), visually represented by the light bands across all metrics. Quantitatively, it underperforms even the constant baseline ( $t_c = 1$ ) in fidelity (FID: 208.30 vs. 199.50). Under the convention  $t : 1 \rightarrow 0$ , this candidate makes  $t_c$  smaller than our rule and therefore keeps strong structural intervention active too

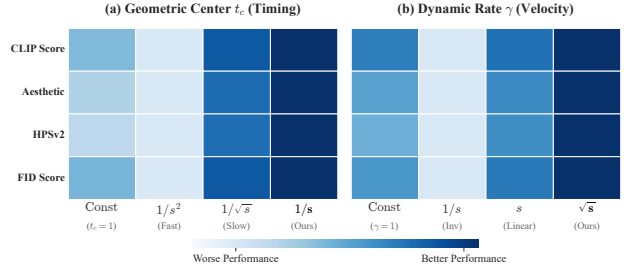


Figure 5. **Performance Landscape of SigMa.** Quantitative analysis on Aesthetic-4K (4096<sup>2</sup>) shows the sensitivity of calibration to parameters. (a) **Timing** ( $t_c$ ): Performance peaks at our inverse-linear timing ( $1/s$ ), consistent with the scale-equivariant center rule. (b) **Rate** ( $\gamma$ ): Our iso-variance scaling ( $\sqrt{s}$ ) achieves the best fidelity; overly gradual ( $1/s$ ) or overly abrupt ( $s$ ) transitions both degrade the structure-to-texture handover.

deep into the clean, texture-dominated stage. The opposite nearby candidate ( $1/\sqrt{s}$ ) switches earlier than our rule and leaves the enlarged layout less supported. In contrast, our **Scale-Equivariant** center law ( $1/s$ )—indicated by the dark vertical column—provides the best tested balance between delayed structural lock-in and late texture recovery, yielding the strongest semantic consistency (CLIP: **29.23**) and the best overall FID (184.73).

**Empirical Support for Iso-Variance Rate** ( $\gamma \propto \sqrt{s}$ ). Figure 5(b) indicates a clear trade-off in transition velocity. The inverse strategy ( $1/s$ ) keeps the mixed regime too wide, allowing structural intervention to bleed into the late texture stage and increasing FID (205.10). The linear strategy ( $s$ ) makes the handover too abrupt. Our **Iso-Variance** scaling ( $\sqrt{s}$ ) strikes the best tested balance by keeping the transition width approximately invariant in standardized feature units, producing the strongest HPSv2 (0.287) and the best FID in this ablation.

### 4.4. Generalization to Extreme Aspect Ratios

Beyond standard square resolutions, we evaluate the robustness of **SigMa** under extreme geometric deformations by generating images at  $4096 \times 1024$  (aspect ratio 4:1). This regime serves as a useful stress test for scale-adaptive calibration: when sequence length increases disproportionately



along a single axis, scale-agnostic interpolation methods may fail to account for asymmetric spectral shifts between the elongated horizon and the base vertical height. Such mismatch can lead to geometric repetition, such as “copy-pasted” mountain peaks in panoramic views. In contrast, SigMa’s axis-decoupled formulation allows independent calibration of modulation timing and rate for height and width, helping synchronize the denoising trajectory with the specific spectral properties of each dimension.

Table 3 compares performance in the 4:1 panoramic regime, exposing a distinct hierarchy among modulation paradigms. Static modulation (YaRN) has the lowest structural fidelity (HPSv2: 0.208), consistent with its scale-agnostic treatment of geometric anisotropy. Rigid dynamic modulation (DyPE) improves global consistency (HPSv2: 0.231) but at the cost of texture degradation (Aesthetic: 5.85). This trend is consistent with the interpretation that a uniform decay across both axes can over-regularize the native vertical dimension (where  $s_h = 1$ ), leading to over-smoothed outputs. **SigMa (Ours)** achieves strong preference scores with axis-decoupled calibration. By synchronizing transition timing and velocity for each dimension, SigMa is better aligned with the anisotropic evolution gap in this stress test, preserving global structural integrity while retaining local textural richness.

Table 3. **Evaluation on Extreme Aspect Ratios** ( $4096 \times 1024$ ). This 4:1 regime stress-tests the “geometric repetition” failure mode of scalar-based methods. See Appendix C.3 for visual evidence.

| Method              | CLIP Score $\uparrow$ | Aesthetic $\uparrow$ | HPSv2 $\uparrow$ |
|---------------------|-----------------------|----------------------|------------------|
| YaRN                | 26.15                 | 5.42                 | 0.208            |
| DyPE-YaRN           | 26.92                 | 5.85                 | 0.231            |
| <b>SigMa (Ours)</b> | <b>27.45</b>          | <b>6.28</b>          | <b>0.254</b>     |

## 5. Conclusion

In this work, we introduced **SigMa**, a training-free framework designed to reduce the structural instability of Diffusion Transformers at ultra-high resolutions. Motivated by the **Evolution Gap**, SigMa adapts positional intervention to the scale-dependent delay of structural lock-in without retraining. Through its two core mechanisms—*Decoupled Geometric Center Alignment* and *Iso-Variance Rate Adaptation*—our method mitigates persistent extrapolation pathologies such as geometric repetition and textural degradation. Extensive evaluations show that SigMa’s axis-decoupled calibration improves robustness across extreme scales (up to  $4K^2$ ) and challenging panoramic aspect ratios. By synchronizing the modulation schedule with the enlarged geometric support of high-resolution generation, SigMa provides a practical foundation for scalable high-fidelity synthesis.

## Impact Statement

**SigMa** makes high-resolution image synthesis more accessible by enabling 4K generation on pre-trained backbones without retraining. This training-free approach can reduce the computational costs and carbon footprint associated with large-scale model scaling, lowering the barrier for creative applications. However, we acknowledge the potential for misuse. The enhanced photorealism at 16-megapixel scale may exacerbate challenges in identifying deepfakes and digital misinformation. Furthermore, SigMa inherits the inherent biases of its underlying pre-trained models. We advocate for the continued development of digital forensics and provenance watermarking to ensure the ethical deployment of high-fidelity generative technologies.

## Acknowledgements

This work was supported by the National Natural Science Foundation of China under Grant 62471394. We would also like to thank the anonymous reviewers for their constructive comments that helped improve the quality of this paper.

## References

- bloc97. Ntk-aware scaled rope allows llama models to have extended (8k+) context size without any fine-tuning and with minimal perplexity degradation. [https://www.reddit.com/r/LocalLLaMA/comments/141z7j5/ntkaware\\_scaled\\_rope\\_allows\\_llama\\_models\\_to\\_have/](https://www.reddit.com/r/LocalLLaMA/comments/141z7j5/ntkaware_scaled_rope_allows_llama_models_to_have/), 2023.
- Chen, J., Yu, J., Ge, C., Yao, L., Xie, E., Wang, Z., Kwok, J., Luo, P., Lu, H., and Li, Z. PixArt- $\alpha$ : Fast training of diffusion transformer for photorealistic text-to-image synthesis. In *International conference on learning representations*, volume 2024, pp. 57611–57640, 2024.
- Chen, S., Wong, S., Chen, L., and Tian, Y. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023.
- Choi, J., Lee, J., Shin, C., Kim, S., Kim, H., and Yoon, S. Perception prioritized training of diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11472–11481, 2022.
- Debnath, L. and Bhatta, D. *Integral transforms and their applications*. Chapman and Hall/CRC, 2016.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171–4186, 2019.

- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024.
- Heo, B., Park, S., Han, D., and Yun, S. Rotary position embedding for vision transformer. In *European Conference on Computer Vision*, pp. 289–305. Springer, 2024.
- Hessel, J., Holtzman, A., Forbes, M., Bras, R. L., and Choi, Y. Clipscore: A reference-free evaluation metric for image captioning, 2022. URL <https://arxiv.org/abs/2104.08718>.
- Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., and Hochreiter, S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Hoogetboom, E., Heek, J., and Salimans, T. simple diffusion: End-to-end diffusion for high resolution images. In *International Conference on Machine Learning*, pp. 13213–13232. PMLR, 2023.
- Issachar, N., Yariv, G., Benaim, S., Adi, Y., Lischinski, D., and Fattal, R. Dype: Dynamic position extrapolation for ultra high resolution diffusion. *arXiv preprint arXiv:2510.20766*, 2025.
- Karras, T., Aittala, M., Aila, T., and Laine, S. Elucidating the design space of diffusion-based generative models. *Advances in neural information processing systems*, 35: 26565–26577, 2022.
- Labs, B. F. Flux. <https://github.com/black-forest-labs/flux>, 2024.
- LeCun, Y. and Bengio, Y. Convolutional networks for images, speech, and time series. *The handbook of brain theory and neural networks*, 1998.
- Lin, S., Liu, B., Li, J., and Yang, X. Common diffusion noise schedules and sample steps are flawed. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 5404–5411, 2024.
- Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 4195–4205, 2023.
- Peng, B., Quesnelle, J., Fan, H., and Shippole, E. Yarn: Efficient context window extension of large language models. In *International Conference on Learning Representations*, volume 2024, pp. 31932–31951, 2024.
- Press, O., Smith, N. A., and Lewis, M. Train short, test long: Attention with linear biases enables input length extrapolation. *arXiv preprint arXiv:2108.12409*, 2021.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language models are unsupervised multitask learners. 2019. URL <https://api.semanticscholar.org/CorpusID:160025533>.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10684–10695, 2022.
- Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E. L., Ghasemipour, K., Gontijo Lopes, R., Karagol Ayan, B., Salimans, T., et al. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, 35: 36479–36494, 2022.
- Schuhmann, C., Beaumont, R., Vencu, R., Gordon, C., Wightman, R., Cherti, M., Coombes, T., Katta, A., Mullis, C., Wortsman, M., et al. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems*, 35: 25278–25294, 2022.
- Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., and Liu, Y. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Wu, C., Li, J., Zhou, J., Lin, J., Gao, K., Yan, K., Yin, S.-m., Bai, S., Xu, X., Chen, Y., et al. Qwen-image technical report. *arXiv preprint arXiv:2508.02324*, 2025.
- Wu, X., Hao, Y., Sun, K., Chen, Y., Zhu, F., Zhao, R., and Li, H. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*, 2023.

Xiao, G., Tian, Y., Chen, B., Han, S., and Lewis, M. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations*, volume 2024, pp. 21875–21895, 2024.

Zhang, J., Huang, Q., Liu, J., Guo, X., and Huang, D. Diffusion-4k: Ultra-high-resolution image synthesis with latent diffusion models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 23464–23473, 2025.



## A. Design Motivation and Empirical Support for the Iso-Variance Law

This appendix complements Mechanism II in Sec. 3.2. We explain why  $\gamma(s_d) = \sqrt{s_d}$  is a natural design rule for **SigMa** and validate it through internal measurements. The derivation combines a standard finite-correlation aggregation model with a direct check of hidden-state statistics inside `FLUX.1-dev`. Together with the ablations and Qwen-Image transfer study, this evidence supports the square-root rate as a scale-adaptive design principle for DiT-based RoPE extrapolation, rather than a backbone-specific tuning choice.

**1. Problem Formulation (Axis-Decoupled).** Let the modulation function for a specific extrapolated spatial axis  $d \in \{w, h\}$  follow the log-odds sigmoid gate in Eq. 9. As the resolution scale  $s_d$  along this axis increases, the number of spatial positions participating in the structure-to-texture handover also increases. The role of  $\gamma_d$  is to control the width of this mixed regime. Our goal is to choose  $\gamma(s_d)$  so that the transition has comparable sharpness across resolutions when measured in standardized feature units rather than raw timestep units.

**2. Spatial Variance Propagation with Finite Correlation.** Modern DiTs typically employ Axial RoPE, which motivates analyzing the elongated height and width dimensions separately to first order. Let  $z_d$  denote an aggregated hidden response along axis  $d$  during the crossover window:

$$z_d = \sum_{i=1}^{n_d} a_i, \quad n_d = s_d n_0, \quad (13)$$

where each  $a_i$  is a local contribution with finite variance. For finite-correlation spatial responses, the covariance sum grows linearly with the number of participating positions:

$$\text{Var}(z_d) = \sum_{i=1}^{n_d} \sum_{j=1}^{n_d} \text{Cov}(a_i, a_j) = O(n_d). \quad (14)$$

The resulting standard deviation therefore scales as

$$\Sigma_{\text{signal}}(s_d) \propto \sqrt{n_d} \propto \sqrt{s_d} \cdot \Sigma_{\text{base}}. \quad (15)$$

This is the same square-root scaling predicted by central-limit-style aggregation when the effective correlation length does not grow proportionally with the full axis length.

**3. The Iso-Variance Criterion.** The sharpness of the phase transition is characterized by the effective width of the sigmoid window,  $\Delta\tau_{\text{window}}$ , representing the interval in log-odds time where the model operates in a mixed spectral state. For the sigmoid in Eq. 9, the local width of the transition band around  $t_c$  is inversely proportional to the slope parameter:  $\Delta\tau_{\text{window}} \propto 1/\gamma$ . To keep the crossover equally sharp across scales in standardized units, the product of the window width and the signal spread is kept approximately invariant:

$$\Delta\tau_{\text{window}} \cdot \Sigma_{\text{signal}}(s_d) = C \quad (\text{Constant}). \quad (16)$$

Substituting Eq. 15 and  $\Delta\tau_{\text{window}} \propto 1/\gamma(s_d)$  yields

$$\frac{1}{\gamma(s_d)} \cdot \sqrt{s_d} \approx C, \quad (17)$$

so

$$\gamma(s_d) \propto \sqrt{s_d}. \quad (18)$$

Thus, the finite-correlation scaling and iso-variance criterion select the square-root law as the rate schedule that keeps the handover width invariant after accounting for the growth of aggregated feature variance.

**4. Empirical Support for the Iso-Variance Law.** We verify this scaling in practice by directly probing the spatial variance propagation inside `FLUX.1-dev` (Labs, 2024). Specifically, we extract Step-4 hidden states, average them across all attention heads in the middle layers, and evaluate 100 prompts from Aesthetic-4K. We fix the height at 1024 and progressively increase the width from 1024 to 4096, thereby isolating the scale factor  $s_w$  along a single axis. For each resolution, we compute the normalized empirical feature standard deviation relative to the  $1024 \times 1024$  baseline. As shown in Table 4, the empirical trajectory closely follows the theoretical multiplier  $\sqrt{s_w}$ , indicating that the finite-correlation model captures the dominant scale trend that SigMa uses for rate adaptation.

Table 4. **Empirical Support for the Iso-Variance Scaling Law.** We extract Step-4 hidden states from FLUX.1-dev, average them over all attention heads in the middle layers, and evaluate 100 Aesthetic-4K prompts. The height is fixed at 1024 while the width scales from 1024 to 4096. The normalized empirical feature standard deviation closely matches the theoretical  $\sqrt{s_w}$  multiplier.

| Scale Factor ( $s_w$ ) | Target Resolution | Theoretical Multiplier ( $\sqrt{s_w}$ ) | Normalized Empirical Feature STD |
|------------------------|-------------------|-----------------------------------------|----------------------------------|
| 1×                     | 1024 × 1024       | 1.000 = $1 \times \sqrt{1}$             | 1.000                            |
| 2×                     | 2048 × 1024       | 1.414 = $1 \times \sqrt{2}$             | 1.398                            |
| 3×                     | 3072 × 1024       | 1.732 = $1 \times \sqrt{3}$             | 1.715                            |
| 4×                     | 4096 × 1024       | 2.000 = $1 \times \sqrt{4}$             | 1.972                            |

**5. Conclusion.** In **SigMa**, we set the proportionality constant to 1, yielding  $\gamma(s_d) = \sqrt{s_d}$ . Together, the variance-scaling argument above, the internal-feature measurements in Table 4, the ablations in Fig. 5(b), and the Qwen-Image transfer results in Appendix C.2 support this formulation as a portable design rule for preserving a consistent structure-to-texture handover across varying aspect ratios and DiT backbones.

## B. SigMa Integration Details

In this section, we detail the mathematical implementation of the **SigMa** framework into specific positional encoding baselines. As outlined in Section 3.3, our integration strategy relies on **Dynamic Parameter Injection**, where the axis-specific spectral gate  $\mu_d(t)$  modulates the baseline’s control parameters (scale factors or frequency thresholds) to synchronize with the diffusion phase.

**Note on Axis-Decoupled Calibration.** Since modern DiTs (e.g., FLUX) utilize Axial RoPE, all operations described below are applied independently to the height and width dimensions. For an extrapolated spatial axis  $d \in \{w, h\}$ , the time-varying effective scale is calculated as  $\hat{s}_d(t) = s_d^{\mu_d(t)}$ , where  $\mu_d(t)$  follows the axis-specific log-odds sigmoid schedule in Eq. 9 with the design laws in Eqs. 11 and 12. If  $s_d = 1$ , the axis is already native and  $\hat{s}_d(t) = 1$ . To avoid overloading notation, we use  $k$  to index RoPE frequency channels inside each axis.

**Integration I: SigMa-PI (Dynamic Scale Injection).** Standard Position Interpolation (PI) applies uniform coordinate downscaling. We render it phase-aware by replacing the static scale factor with the time-varying effective scale  $\hat{s}_d(t)$ :

$$g_d(m_d, t) = \frac{m_d}{\hat{s}_d(t)}, \quad h_d(\theta_k, t) = \theta_k. \quad (19)$$

*Mechanism:* Early in the sampling process (Structure Phase), larger  $\mu_d(t)$  makes  $\hat{s}_d(t)$  close to the target scale and strengthens global structural constraints via interpolation. As the process transitions to the Texture Phase,  $\mu_d(t)$  decreases, reducing the effective scale and relaxing the mapping toward the native coordinate system. This gradually restores high-frequency response for sharp detail generation once the global structure is more stable.

**Integration II: SigMa-NTK (Dynamic Frequency Limit).** NTK-aware interpolation prevents aliasing by rescaling high frequencies while preserving low frequencies. We generalize this by modulating the limit frequency via the axis-specific effective scale:

$$g_d(m_d, t) = m_d, \quad h_d(\theta_k, t) = \frac{\theta_k}{\hat{s}_d(t)^{\frac{2k}{D-2}}}. \quad (20)$$

*Mechanism:* The interpolation strength is dynamic. Low frequencies are strongly interpolated initially to secure the layout. As  $\mu_d(t)$  decays, the effective scale reduces, gradually expanding the model’s bandwidth to capture high-frequency textures that would otherwise be dampened by static interpolation.

**Integration III: SigMa-YaRN (Dynamic Boundary Gating).** YaRN combines NTK-by-parts frequency scaling with a ramp function  $\rho(r)$  to mix interpolation and extrapolation. Unlike PI/NTK, we integrate SigMa by modulating the *ramp boundaries* rather than a global scale. We define the time-varying ramp function  $\rho_d(r, t)$  by scaling the original thresholds

Table 5. **Additional Quantitative Evaluation.** Complementing the main text, we provide results for **Portrait Extrapolation** ( $2048 \times 3072$ ) and intermediate  $3 \times$  **Square Super-Resolution** ( $3072^2$ ). Consistent with the main results, **SigMa** maintains strong overall performance across both additional geometries. **Bold** indicates best in group.

| Method                    | Portrait Extrapolation ( $2048 \times 3072$ ) |                |                |                 |                |                |                  | Square Extrapolation ( $3 \times$ Scale, $3072^2$ ) |                |                |                 |                |                |                  |
|---------------------------|-----------------------------------------------|----------------|----------------|-----------------|----------------|----------------|------------------|-----------------------------------------------------|----------------|----------------|-----------------|----------------|----------------|------------------|
|                           | DrawBench                                     |                |                | Aesthetic-4K    |                |                |                  | DrawBench                                           |                |                | Aesthetic-4K    |                |                |                  |
|                           | CLIP $\uparrow$                               | Aes $\uparrow$ | HPS $\uparrow$ | CLIP $\uparrow$ | Aes $\uparrow$ | HPS $\uparrow$ | FID $\downarrow$ | CLIP $\uparrow$                                     | Aes $\uparrow$ | HPS $\uparrow$ | CLIP $\uparrow$ | Aes $\uparrow$ | HPS $\uparrow$ | FID $\downarrow$ |
| <i>Baseline</i>           |                                               |                |                |                 |                |                |                  |                                                     |                |                |                 |                |                |                  |
| FLUX.1-dev                | 26.58                                         | 5.63           | 0.253          | 28.74           | 6.33           | 0.262          | 188.45           | 24.98                                               | 4.99           | 0.217          | 27.92           | 6.16           | 0.253          | 189.21           |
| <i>NTK-based Methods</i>  |                                               |                |                |                 |                |                |                  |                                                     |                |                |                 |                |                |                  |
| NTK-aware                 | 27.45                                         | 5.54           | 0.260          | 29.17           | 6.60           | 0.273          | 186.98           | 25.73                                               | 5.20           | 0.231          | 28.71           | 6.42           | 0.271          | 184.72           |
| DyPE-NTK                  | 27.42                                         | 5.60           | 0.265          | 28.94           | 6.66           | 0.275          | 181.23           | 26.38                                               | 5.42           | 0.248          | 28.43           | 6.51           | 0.278          | 180.18           |
| <b>SigMa-NTK</b>          | <b>27.77</b>                                  | <b>5.63</b>    | <b>0.266</b>   | <b>29.20</b>    | <b>6.73</b>    | <b>0.281</b>   | <b>179.92</b>    | <b>26.41</b>                                        | <b>5.45</b>    | <b>0.249</b>   | <b>28.47</b>    | <b>6.57</b>    | <b>0.279</b>   | <b>179.13</b>    |
| <i>YaRN-based Methods</i> |                                               |                |                |                 |                |                |                  |                                                     |                |                |                 |                |                |                  |
| YaRN                      | 27.47                                         | 5.72           | 0.270          | 28.95           | 6.72           | 0.280          | 181.12           | 27.35                                               | 5.49           | 0.255          | 29.02           | 6.62           | 0.283          | 183.93           |
| DyPE-YaRN                 | 27.92                                         | 5.81           | 0.277          | 29.26           | 6.76           | 0.282          | <b>180.11</b>    | 27.77                                               | 5.60           | 0.267          | 29.32           | 6.69           | <b>0.297</b>   | 182.71           |
| <b>SigMa-YaRN</b>         | <b>27.96</b>                                  | <b>5.85</b>    | <b>0.279</b>   | <b>29.31</b>    | <b>6.84</b>    | <b>0.285</b>   | 181.27           | <b>28.05</b>                                        | <b>5.62</b>    | <b>0.268</b>   | <b>29.48</b>    | <b>6.78</b>    | 0.294          | <b>181.88</b>    |

$(\beta_{min}, \beta_{max})$  with  $\mu_d(t)$ :

$$\rho_d(r, t) = \begin{cases} 0, & r < \beta_{min} \cdot \mu_d(t) \\ \frac{r - \beta_{min} \cdot \mu_d(t)}{\beta_{max} \cdot \mu_d(t) - \beta_{min} \cdot \mu_d(t)}, & \beta_{min} \cdot \mu_d(t) \leq r \leq \beta_{max} \cdot \mu_d(t) \\ 1, & r > \beta_{max} \cdot \mu_d(t) \end{cases} \quad (21)$$

where  $r = L_d / \lambda_k$ . The corresponding frequency response is

$$h_d(\theta_k, t) = (1 - \rho_d(r, t)) \frac{\theta_k}{s_d} + \rho_d(r, t) \theta_k. \quad (22)$$

*Mechanism:* As  $\mu_d(t)$  decreases during sampling, the effective thresholds shift towards zero on each axis. Larger gate values keep the ramp closer to the original YaRN boundaries and preserve structural consistency, while smaller values shrink the thresholds and move the response toward native RoPE for positive wavelength ratios. Because the gate remains in  $(0, 1)$  on the open interval  $t \in (0, 1)$ , the denominator is well-defined when  $\beta_{max} > \beta_{min}$ . Thus the positional intervention is smoothly relaxed as denoising moves from structure formation to texture refinement, without requiring output normalization or any additional compression parameter.

## C. Additional Experimental Results

### C.1. Additional Quantitative Results

This section complements the main-text experiments by adding two settings not shown in Table 1: a portrait extrapolation regime ( $2048 \times 3072$ ) and an intermediate  $3 \times$  square regime ( $3072^2$ ). Table 5 shows that the same trend persists away from the headline settings in the main paper: SigMa remains consistently among the strongest extrapolators across these additional geometries, especially when compared with static baselines at larger spatial spans.

### C.2. Qwen-Image Generalization Results

To provide preliminary evidence that **SigMa** is not specific to FLUX, we additionally evaluate it on **Qwen-Image** (Wu et al., 2025), a stronger contemporary text-to-image DiT backbone. Table 6 complements the FLUX.1-dev results in the main text by repeating the same extrapolation protocol on a second backbone: starting from the  $1024 \times 1024$  base resolution, we generate  $4096 \times 4096$  images on **Aesthetic-4K** and report CLIP, Aesthetic Score, HPSv2, and FID.

These results provide initial transfer evidence: **SigMa** improves all reported metrics over the native Qwen-Image baseline and yields small but consistent gains over the competing extrapolation methods in this setting.

Table 6. **Evaluation on Qwen-Image** ( $4096 \times 4096$ ). Starting from the  $1024 \times 1024$  base resolution, we extrapolate to  $4096 \times 4096$  on Aesthetic-4K and compare the native Qwen-Image baseline, YaRN, DyPE-YaRN, and SigMa-YaRN.

| Method                   | CLIP $\uparrow$ | Aes. $\uparrow$ | HPSv2 $\uparrow$ | FID $\downarrow$ |
|--------------------------|-----------------|-----------------|------------------|------------------|
| Qwen-Image baseline      | 27.42           | 5.61            | 0.212            | 200.73           |
| YaRN                     | 28.73           | 6.24            | 0.239            | 198.67           |
| DyPE-YaRN                | 28.81           | 6.33            | 0.277            | 197.01           |
| <b>SigMa-YaRN (Ours)</b> | <b>28.93</b>    | <b>6.36</b>     | <b>0.281</b>     | <b>196.69</b>    |

### C.3. Qualitative Results on Extreme Aspect Ratios

In Figure 6, we provide visual evidence supporting the quantitative results in Section 4.4.

"A Moai statue gazes upward against a starry night sky, with a colorful sunset illuminating the horizon."



YaRN



DyPE-YaRN



SigMa-YaRN

Figure 6. **Reduction of Geometric Repetition in Panoramic Generation** ( $4096 \times 1024$ ). The input prompt specifies “A Moai statue...” (singular). **Top/Middle:** Due to scale-agnostic global interpolation, both **YaRN** and **DyPE** fail to maintain the global field of view across the 4 : 1 aspect ratio, resulting in *Geometric Repetition* (hallucinating a second statue) to fill the widened canvas. **Bottom: SigMa**, equipped with **Axis-Decoupled Calibration**, maintains a single, coherent subject in this example. By applying the scale-equivariant center law along the width axis ( $t_c = 1/s_w$ ), the result is consistent with our scale-aware interpretation of anisotropic generation.